

Research Idea: PJ231356.6

ALMA Thesis Planner (automated pipeline)

Title: How compact is the dust? A uv-distance flux-decomposition and peak brightness-temperature census of the public 2022.1.01311.S continuum data

The question. Project 2022.1.01311.S was built around a single physical question: does star formation in extreme dusty star-forming galaxies (DSFGs) saturate at an Eddington-like ceiling in star formation surface density? Whether it does depends entirely on *how concentrated the dust emission actually is* — a given far-IR luminosity spread over a few kpc is unremarkable, and the same luminosity packed into 100–500 pc is the regime the proposal was written to probe. This thesis measures exactly that concentration, and only that, for every public target in the program, using two observables that require no lens model, no source-plane reconstruction, and no assumption about what shape the emission has.

Observable 1 — flux versus angular scale, straight from the visibilities. For each target, azimuthally bin the calibrated continuum visibilities in uv distance and report three numbers: (a) the total flux density from the short-baseline plateau; (b) the flux density surviving on the longest well-sampled baselines, i.e. the *compact fraction*; and (c) the uv distance at which the binned amplitude falls to half the plateau, which is a characteristic angular scale of the emission. These are morphology-agnostic — they are statements about how much flux lives on which angular scale, not about whether the source is a Gaussian, a clump, or an arc — and their uncertainties come directly from the visibility weights, which are near-independent per visibility, unlike correlated CLEAN-image pixels.

Three technical points are treated as first-class parts of the work, not footnotes. (i) Vector-averaged amplitudes are positively biased at low S/N, so the student phase-shifts each dataset to the emission centroid and averages the *real* part of the visibility, quoting the imaginary part as a null/asymmetry check, rather than averaging $|V|$. (ii) The compact fraction depends on where the longest-baseline cut is placed and on each target’s uv coverage and sensitivity, so it is quoted over a *common* uv range shared by the sample as the headline number, with each target’s native-coverage value given alongside — this is what makes the census uniform rather than nominally uniform. (iii) A curved, non-circular source does not make azimuthal binning invalid — the binned profile is a well-defined angle-averaged measure of power on each scale — but it does mean the half-amplitude uv distance is a characteristic scale, not a source size, and it is labeled as such throughout.

Observable 2 — peak rest-frame brightness temperature. From the image-plane products, invert the full Planck function (not the Rayleigh–Jeans approximation: at $870\ \mu\text{m}$ observed and $z \approx 2$ the rest frequency is high enough that RJ fails) to obtain the peak rest-frame brightness temperature per target. Specific intensity divided by ν^3 is a lensing invariant, so this needs no magnification correction. Every value is reported as a *lower* bound on the true peak, because it is degraded by beam filling factor and by optical depth below unity; the CMB at these redshifts is stated as a background/contrast floor rather than ignored. Peak T_B is quoted both at native

resolution and after convolution to a beam common to the sample, so the numbers are comparable across targets with different delivered beams.

What is deliberately not attempted. No parametric component decomposition, no source-plane reconstruction, and no identification of multiply-imaged counterparts. All angular results are reported in the observed frame; converting them to source-plane physical scales requires a lens model and is explicitly left to the collaboration. Differential magnification is named as the reason a compact fraction is an observed-frame quantity, not an intrinsic one.

Data and its stated provenance. Only archival products from 2022.1.01311.S — calibrated measurement sets for Observable 1, imaging products for Observable 2. The abstract quotes the continuum imaging as "870micron/1mm" and does not label these as rest- or observed-frame; they correspond to ALMA receiver bands, so they are treated as observed-frame and verified against the archive's spectral-window metadata before any analysis. The abstract's 0.1" resolution and 100–500 pc scales are *requested/expected* values, not delivered measurements; the delivered synthesized beam, sensitivity, and uv coverage are inventoried per target from the archive. The abstract gives $z = 1\text{--}3$ as a sample range only, and $>10^{13} L_{\odot}$ as an already-magnification-corrected intrinsic luminosity; per-source redshifts are used where project metadata supplies them, and the full $z = 1\text{--}3$ range is carried as an explicit bracket where they do not. The description does not state proprietary status, so archive availability is confirmed in week 1 and the census simply covers whichever targets are public — the deliverable scales cleanly with that number.

Schedule (24 weeks; write-up is a scheduled block, not leftover time). Weeks 1–3: confirm archive availability; inventory delivered beams, sensitivities, and uv coverage; fix the common uv range and the common convolution beam *in writing* before looking at results. Weeks 4–8: build and validate the binning pipeline on the two highest-S/N targets, including injection–recovery through the real uv sampling using a deliberately non-Gaussian input (a curved arc), to confirm the half-amplitude scale and compact fraction are recovered and to calibrate their systematic floor. Weeks 9–16: run the full public sample. Weeks 17–19: brightness-temperature maps and the peak-T_B table. Weeks 20–24: write-up, with all measurements frozen at week 20.

Deliverables. A table, one row per public target: total flux, compact fraction at the common uv cut and at native coverage, characteristic angular scale with injection-calibrated uncertainty, and peak rest-frame T_B at native and common beam with the redshift bracket shown. Plus the binned uv-amplitude profiles themselves as a figure set. Every entry is a real measurement even when the answer is negative: "the emission in target X is dominated by structure smoothed out beyond the common uv cut" is a quantitative constraint on concentration, and "peak rest-frame T_B is only a few K" is a genuine joint bound on filling factor and optical depth. That is what keeps the project finishable and keeps it pointed at the proposal's Eddington-limit question rather than at its lens-modeling goal, which is not thesis-sized.

Explicitly optional, only if the census is frozen on time: a qualitative morphological comparison of one target's continuum against the archival rest-frame optical HST imaging the proposal itself names for the dust-extinction-geometry question. Not part of required scope.